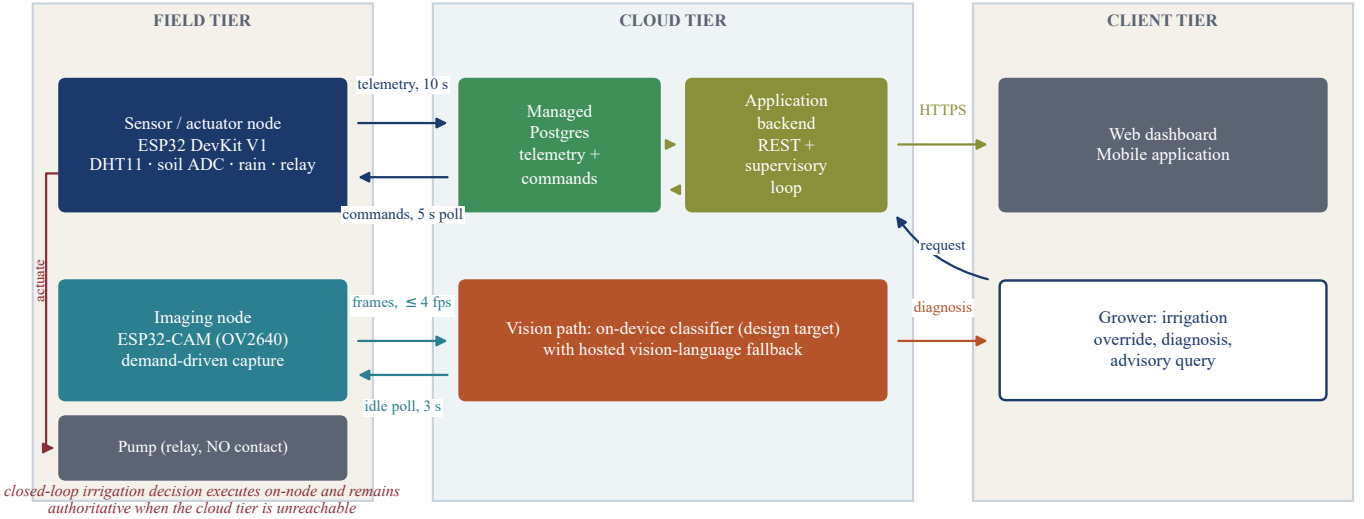

FIELD TIER
CLOUD TIER
CLIENT TIER
Sensor / actuator node
ESP32 DevKit V1
DHT11 · soil ADC · rain · relay
telemetry, 10 s
commands, 5 s poll
Managed
Postgres
telemetry +
commands
Application
backend
REST +
supervisory
loop
HTTPS
Web dashboard
Mobile application
actuate
request
Imaging node
ESP32-CAM (OV2640)
demand-driven capture
frames, ≤ 4 fps
idle poll, 3 s
Vision path: on-device classifier (design target)
with hosted vision-language fallback
diagnosis
Grower: irrigation
override, diagnosis,
advisory query
Pump (relay, NO contact)
closed-loop irrigation decision executes on-node and remains
authoritative when the cloud tier is unreachable